

SYSTEMS IMMUNOLOGY

Assessing AI's cognitive abilities for scientific discovery in the field of systems vaccinology

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The advent of large language models (LLMs) has transformed academic research by accelerating hypothesis generation and data analysis. LLMs can help researchers uncover patterns and insights from vast datasets to foster innovative scientific discovery. However, questions arise regarding the creative capacity of artificial intelligence (AI), especially in biologically complex fields such as vaccinology. This study evaluates the ability of LLMs to generate hypotheses, design experiments, and infer broader biological principles through a proposed framework called “The Creation Game.” Using three case studies—general control nonderepressible 2 (GCN2)’s role in dendritic cell antigen presentation via stress response, sterol regulatory element-binding protein (SREBP)’s influence on metabolic responses, and Toll-like receptor 5 (TLR5)’s connection to microbiota-driven vaccine efficacy—we assessed AI’s accuracy, logic, and creativity. The findings underscore the potential of LLMs to accelerate vaccine research while emphasizing the importance of ethical oversight. By complementing human creativity, AI could potentially transform hypothesis-driven science, paving the way for tailored vaccination strategies and deeper insights into human immunity.

INTRODUCTION

The advent of large language models (LLMs) has considerably transformed the academic landscape, offering researchers tools to enhance their investigative processes. Models such as ChatGPT (OpenAI), Microsoft Copilot (Microsoft Corporation), SciSpace (Typeset.io), and LLaMA (Meta Platforms) are well suited to analyzing and inferring patterns from vast datasets, granting researchers unparalleled access to immunological, molecular, and genomic knowledge and the ability to generate new hypotheses (1–5). By analyzing extensive datasets and synthesizing information from diverse sources, these sophisticated artificial intelligence (AI) systems offer scholars insights that may have previously gone unnoticed (5).

LLMs have already demonstrated their potential for scientific discovery in molecular biology. For example, the LLM4SD (LLMs for Scientific Discovery) framework has successfully leveraged LLMs to synthesize knowledge from literature and infer previously unidentified patterns from molecular datasets to improve molecular property prediction (6). By identifying relationships between molecular structures and their biological functions, LLM4SD has outperformed existing benchmarks, demonstrating how AI can generate interpretable insights (6). This success raises the possibility that similar approaches could be applied to other scientific domains, such as vaccinology, where AI could assist in hypothesis-driven research and mechanistic discovery.

From an immunological perspective, this question is particularly timely and consequential. Immunology—especially systems vaccinology—presents a challenge for LLMs due to the biological complexity, context dependence, and mechanistic depth required to formulate meaningful hypotheses. Systems vaccinology aims to understand how the immune system responds to vaccination by integrating large-scale molecular data, such as transcriptomics, proteomics, and

metabolomics, with cellular and functional immune readouts (7, 8). This holistic approach moves beyond studying single pathways to capture how innate sensors, antigen-presenting cells (APCs), and adaptive B and T cells coordinate protective immunity. Such biological complexity, coupled with strong context dependence across age, environment, and pathogen exposure, presents a challenge for LLMs. Their ability to synthesize these multilayered datasets and relate them to mechanistic processes tests whether LLMs can genuinely contribute to mechanistic immunology and vaccine discovery.

However, because AI continues to develop, an important question emerges: Can machines truly create in the same way humans do (9)? The interplay between AI and human creativity is pivotal in shaping new methodologies for scientific discovery (10). Although the classic Turing test evaluates AI’s ability to mimic human intelligence, concerns have been raised that AI’s ability to generate scientific knowledge may be an illusion of objectivity rather than genuine understanding (11). AI models can synthesize information efficiently, but this efficiency may come at the cost of critical human oversight, potentially reinforcing scientific monocultures where dominant perspectives overshadow alternative methodologies (11).

To explore this issue, we recently proposed the concept of “The Creation Game” as a test of AI’s potential for human-like creativity (9). This framework was designed to challenge AI to generate innovative scientific hypotheses, propose experiments, contextualize potential results of the experiment, and infer broader biological principles from such results in ways that mirror the creativity exhibited by expert human researchers. For example, the field of systems vaccinology, which involves the integration of immunology, genomics, and machine learning to understand and predict vaccine responses, requires the generation of new hypotheses based on the intricate interplay of immune pathways and host factors (7–9). This approach has identified distinct vaccination-induced, as well as baseline, molecular signatures that are associated with and can predict the magnitude of antibody responses after immunization with vaccines against pathogens such as yellow fever, seasonal influenza, shingles, malaria, and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) (12–21). These findings have uncovered previously

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unrecognized biological mechanisms underlying vaccine-induced immunity. In a recent large-scale meta-analysis, encompassing data from more than 3000 samples collected from 820 adults across 28 independent studies covering 13 different vaccines, researchers identified a conserved plasmablast-associated gene expression profile and baseline signatures that reliably predict antibody responses across multiple vaccines (22, 23). In addition, a recent study has defined molecular signatures that predict the durability of the antibody responses to multiple vaccines in humans, as well as some of the underlying immunological mechanisms (24). The immunological complexity of these findings, ranging from innate sensing pathways to adaptive cell differentiation, highlights the cognitive and integrative demands of creative hypothesis generation in this field.

Identifying molecular signatures of vaccine responses and proposing mechanisms for these signatures often require a combination of biological insight, experimental design, and critical evaluation—tasks that are traditionally the realm of human critical thinking (9). The Creation Game allows us to test AI's creative capacity in generating hypotheses about immune responses, such as those that predict vaccine responses. In this context, AI could potentially assist in hypothesizing specific immune pathways, just as human researchers do. However, it remains to be seen whether LLMs can replicate this creative process, particularly in biologically complex fields such as vaccinology.

Given that such analyses have already advanced our mechanistic understanding of human immunity, we proposed testing AI LLMs in The Creation Game with the following example prompt: “We have defined a gene expression signature as a predictor of the antibody response to influenza vaccination. Could you: (i) survey the literature and propose a potential hypothesis to explain this result? (ii) Suggest an experiment to biologically validate the hypothesis? (iii) Contextualize the potential outcomes of this experiment? (iv) Infer broader biological principles from the results of the experiment?”

Such a challenge will test AI's skills with regard to accuracy of recall of relevant literature, deduction, logic, and creativity. Recall looks at whether AI can accurately, and without hallucinations, survey the literature to identify relevant prior publications. Deduction and logic examine whether AI can formulate a potential hypothesis from the gene expression signature and propose experiments to test such a hypothesis. Creativity assesses whether AI can infer broader principles from a result and postulate a concept that regional differences in vaccine efficacy across the world could stem from variations in microbiota. In this manner, The Creation Game could be operationalized by querying LLMs with biologically validated signatures of vaccine efficacy to assess their capacity for accurate recall, hypothesis generation, logic, and creativity.

This framework was not designed to produce implementable experiments per se but to probe whether LLMs can emulate the early stages of scientific reasoning, particularly hypothesis formation, deductive inference, and conceptual abstraction on the basis of existing biological signals. In doing so, we aimed to identify patterns in how LLMs assemble scientific ideas rather than to validate their ability to create truly innovative, actionable insights.

Although The Creation Game encourages grounding in literature, this very constraint highlights a critical limitation: LLMs are trained on the same scientific canon that they are asked to interpret, blurring the line between genuine ideation and high-dimensional recall. To explore this, we tested alternative prompting strategies with ChatGPT-4.5 that deliberately encouraged responses outside

canonical interpretations to better evaluate the generative ceiling of current models. Thus, rather than claiming to measure creativity per se, we present The Creation Game as a conceptual tool to probe the scaffolding of the mechanistic reasoning in LLMs.

In this study, we tested five different LLMs—ChatGPT-4o, ChatGPT-4.5, Microsoft Copilot, LLaMA, and SciSpace—on the basis of a four-tier evaluation framework: (i) accuracy of recall of relevant literature, (ii) formulation of biological hypotheses, (iii) proposal of experiments to validate hypotheses, and (iv) inference of broader conceptual significance of results. We evaluated the effectiveness of various models through the following three distinct case studies in systems vaccinology: (i) The general control nonderepressible 2 (GCN2) case study focused on the role of GCN2 in immunogenicity, emphasizing its function in enhancing antigen presentation in dendritic cells and controlling gut inflammation (12, 25, 26); (ii) the sterol regulatory element-binding protein (SREBP) study examined SREBP's significance in metabolic responses to vaccination and its essential role in effective B cell responses (17, 27); (iii) the Toll-like receptor 5 (TLR5) case study investigated how antibiotic-induced alterations in the gut microbiome affect vaccine immunity, as well as broader implications in vaccinology within high-throughput biology (28, 29). These case studies assess the models' accuracy and their creative thinking capabilities.

Amino acid sensing and metabolic stress play pivotal roles in shaping immune responses. The kinase GCN2 acts as a key sensor of amino acid depletion in mammals and plays a central role in coordinating the integrated stress response. Under conditions of amino acid scarcity, GCN2 modulates protein synthesis by phosphorylating eukaryotic translation initiation factor 2 α (eIF2 α) at the Ser⁵¹ (S51) residue. This phosphorylation event impairs the formation of the active preinitiation complex eIF2–guanosine 5'-triphosphate–methionine transfer RNA (tRNAMet), reducing polysome assembly and leading to the accumulation of stress granules (30). Previous studies have demonstrated that GCN2 has antiviral properties and influences the survival and expansion of T cells (31–33). However, its role in linking innate immune signaling to adaptive immunity remained uncertain. Our previous studies identified GCN2 activation as a correlate of vaccine-induced immunity, showing that viral stimulation of this pathway promotes autophagy in dendritic cells and enhances antigen presentation to both CD4⁺ and CD8⁺ T cells after vaccination with the live-attenuated yellow fever 17D (YF-17D) vaccine, which was derived from the 17D strain of yellow fever virus (12, 25, 26). Given that the expression of GCN2 was a correlate of the CD8 T cell response to YF-17D (12), we have investigated the mechanisms by which GCN2 regulates antigen-specific T cell responses after immunization with YF-17D in mice. Our findings revealed a crucial role for virus-triggered activation of GCN2 in directing dendritic cells to engage autophagy, thereby enhancing antigen presentation to both CD4 and CD8 T cells. These results uncovered a previously unrecognized connection between the virus-induced integrated stress response in dendritic cells and the activation of the adaptive immune system. This established link between cellular metabolism, stress signaling, and adaptive immunity provides an ideal context to test the reasoning capacity of LLMs. This established link between nutrient sensing, stress signaling, and adaptive immunity provides an ideal context to test whether LLMs can recognize complex mechanistic relationships underlying immune regulation and vaccine responsiveness.

Sterol metabolism similarly integrates cellular function with immune activity. It plays a crucial role in human health and is a key

target for cardiovascular disease therapies. The SREBP pathway governs lipid biosynthesis and is the molecular target of statin therapy widely used to prevent cardiovascular disease. Statins, widely prescribed in the United States, act by inhibiting 3-hydroxy-3-methylglutaryl-coenzyme A reductase (HMGCR), a key enzyme in the cholesterol biosynthesis pathway regulated by SREBP signaling. Statin use has been linked to weaker antibody responses after influenza vaccination, and individuals on these medications may have a higher risk of severe respiratory illness from influenza despite being vaccinated (34, 35). Our previous study provides a potential explanation for this phenomenon, suggesting that statins may impair B cell metabolism, thereby diminishing vaccine-induced immunity (27). These findings highlight the critical intersection between metabolic regulation and humoral responses, positioning SREBP as an informative model for evaluating whether LLMs can infer connections between lipid metabolism, immune activation, and variability in vaccine efficacy.

The microbiota also plays a crucial role in immune maturation and vaccine responsiveness. TLR5, which recognizes bacterial flagellin, serves as a key molecular bridge between host-microbiota interactions and innate immune activation. Studies have shown that antibiotic-mediated depletion of commensal bacteria impairs TLR5 signaling, resulting in diminished antibody responses to vaccination (28, 29). Our previous findings and others have underscored how microbial diversity and pattern recognition receptor signaling influence vaccine efficacy. This makes TLR5 a compelling case study to test whether LLMs can integrate environmental and immunological information to predict vaccine responses.

Together, these three case studies represent complementary dimensions of immune regulation: metabolic stress signaling (GCN2), lipid metabolism (SREBP), and host-microbiota interaction (TLR5). Each captures a distinct layer of systems vaccinology—from intracellular signaling to organismal and environmental influences—that collectively define vaccine responsiveness. By applying the same four-tier framework across these diverse mechanisms, we aimed to evaluate whether LLMs can generalize their reasoning beyond factual recall to mechanistic interpretation, experimental design, and conceptual synthesis. This multicontext approach provides a rigorous test of AI's potential to contribute meaningfully to hypothesis generation and systems-level understanding in vaccinology.

RESULTS

Evaluating the cognitive potential of LLMs in systems vaccinology

We evaluated the cognitive abilities of five LLMs—ChatGPT-4o, ChatGPT-4.5, Microsoft Copilot, LLaMA, and SciSpace—using a structured framework to analyze molecular signatures previously identified in systems vaccinology studies as predictors of vaccine-induced immune responses. These signatures, derived from multiomics profiling, capture baseline and early postvaccination transcriptional patterns linked to immune magnitude and durability (7, 18). The objective was to assess each model's ability to interpret these signatures by (i) formulating hypotheses about mechanisms of immune regulation, (ii) proposing experiments to biologically validate these signatures, (iii) interpreting the potential results from such experiments, and (iv) inferring broader biological principles from such experiments.

To conduct our evaluation, we formulated a thorough query for each of the three case studies: GCN2, SREBP, and TLR5. Each case

study was designed to investigate specific molecular pathways and their implications for immune responses and vaccine efficacy. We input these queries into each of the five LLMs and compared their outputs against historical data and previously conducted experiments outlined in the introduction. This comparison allowed us to evaluate the models' ability to synthesize information and generate contextually relevant insights (Fig. 1).

Our four-tier evaluation framework included assessing the broader conceptual importance of the results, the formulation of experiments to validate hypotheses, the formulation of biological hypotheses, and accurate data mining. By applying this framework, we aimed to determine how well these LLMs could think and contribute to scholarly inquiry in the field of immunology. The results of this evaluation will provide valuable insights into the strengths and limitations of LLMs in generating hypotheses and facilitating research discussions, ultimately contributing to the ongoing dialogue about the role of AI in advancing scientific research.

GCN2 as a link between the integrated stress response and adaptive immunity in yellow fever vaccination

Our previous systems vaccinology approach to studying the immune response to the yellow fever vaccine in humans uncovered early blood transcriptional signatures that predicted the strength of the subsequent CD8 T cell response to vaccination with the yellow fever vaccine YF-17D (12). Among the genes identified within these predictive signatures was the GCN2 kinase gene, also known as *EIF2AK4*.

In that work, GCN2 was shown to activate autophagy in dendritic cells, enhancing antigen presentation and connecting nutrient-sensing stress pathways to adaptive immunity (Fig. 2) (25, 26). In the current study, this established mechanism served as a benchmark for testing LLM reasoning, allowing us to evaluate whether AI systems could recognize and expand upon known links between metabolic stress, antigen processing, and T cell priming.

Using this previously validated model, we assessed each LLM's ability to (i) accurately recall the literature and molecular details of GCN2 signaling, (ii) formulate hypotheses connecting amino acid sensing to vaccine-induced immune activation, (iii) propose experimental approaches to validate such mechanisms, and (iv) infer broader conceptual insights into how metabolic stress pathways regulate antigen presentation and T cell responses. Framing GCN2 as a case study enabled us to probe whether LLMs can generalize from a well-defined molecular pathway to more abstract principles connecting metabolism and immunity.

Evaluating GCN2 analysis: Strengths and limitations of LLMs in immunology research

The evaluation of GCN2 across the five LLMs—ChatGPT-4o, ChatGPT-4.5, Microsoft Copilot, SciSpace, and LLaMA—revealed a nuanced landscape of capabilities relevant to data mining, hypothesis formulation, experimental design, and conceptual significance in the context of immunology research. ChatGPT-4o, Microsoft Copilot, and SciSpace demonstrated strong proficiency in identifying relevant literature and extracting key data points related to GCN2 functions, each achieving 100% validity, and in ability to retrieve existing and appropriate literature, based on the reference dataset (table S1), as illustrated in Fig. 3A. Although LLaMA showed lower validity, it still contributed relevant albeit less consistent information. The more detailed literature synthesis and biological interpretation were generated by ChatGPT-4.5 (Fig. 3, A to D), showcasing its

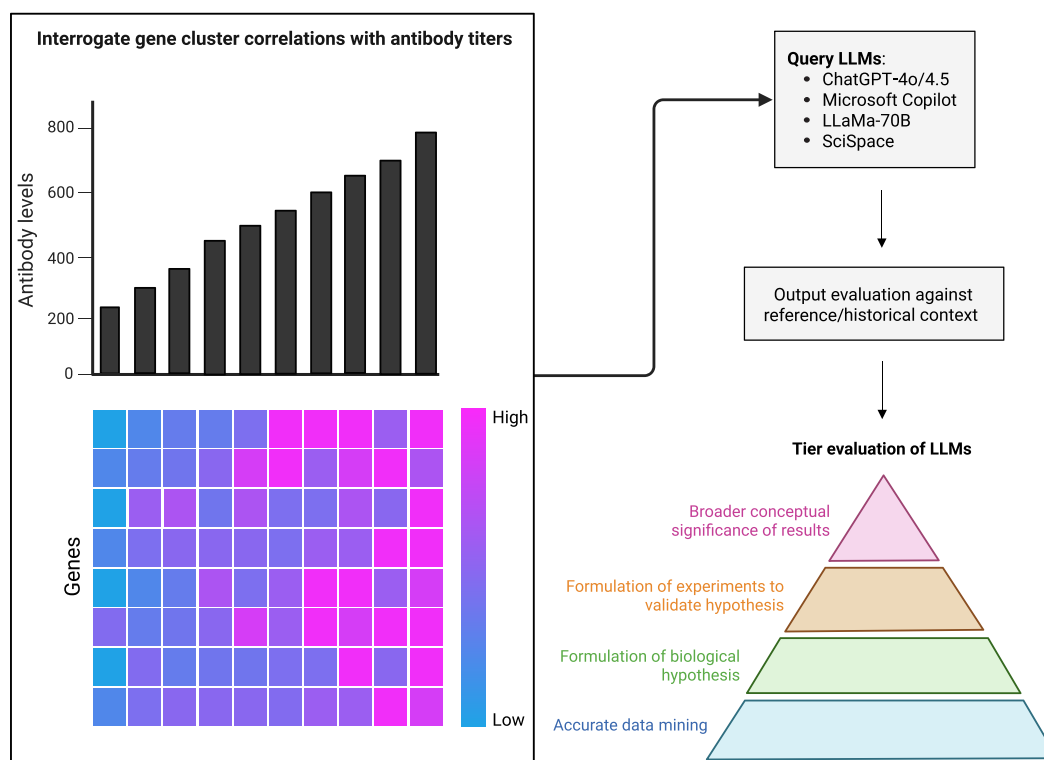


Fig. 1. Overview of LLM evaluation framework for scientific discovery. Gene expression cluster correlations with antibody titers found in experimental results from previous vaccine studies can be used to query LLMs (left). Historical experimental data that underwent analysis and interpretation by humans are compared with LLMs' ability to "think" and arrive at similar conclusions as well as provide further interpretation and experimental recommendations. Queries for three different case studies were evaluated with five different LLMs, and the output of each LLM was assessed on the basis of a four-tier evaluation system (right).

ability to contextualize GCN2 expression with immunometabolic pathways and CD8⁺ T cell responses.

In terms of hypothesis generation, all models produced biologically plausible ideas related to GCN2's role in immune function. ChatGPT-4o, Microsoft Copilot, and SciSpace offered coherent and innovative hypotheses (Fig. 3B), but ChatGPT-4.5, whose creativity assessment outputs are detailed in fig. S1A, provided the most mechanistically diverse and immunologically relevant set of hypotheses. These included roles for GCN2 in metabolic reprogramming, innate immune epigenetic imprinting, selective translation of immune modulators, inflammasome activation, extracellular vesicle signaling, and microbiome interactions. LLaMA's hypotheses, in comparison, were broader and less directly applicable to experimental validation.

When considering experimental design capabilities, the top three models outlined coherent and feasible experimental approaches (Fig. 3C). ChatGPT-4.5 went further by proposing a detailed and technically robust pipeline for investigating GCN2-driven extracellular vesicle signaling, encompassing peripheral blood mononuclear cell (PBMC) culture, extracellular vesicle (EV) isolation, multiomics cargo profiling, and in vivo validation, summarized in fig. S1B. In contrast, LLaMA's experimental designs were often biologically unrealistic or lacked necessary methodological detail.

Last, regarding conceptual importance, ChatGPT-4.5 provided the strongest and most integrated framing of GCN2's immunological roles, including translational perspectives such as using EV cargo signatures as biomarkers of vaccine efficacy or potential therapeutic vectors (Fig. 3D and fig. S1B). ChatGPT-4o, Microsoft Copilot, and SciSpace also contributed contextually relevant insights, whereas

LLaMA's contributions were limited in depth and specificity. Overall, this comparative evaluation highlights the importance of leveraging advanced LLMs, such as ChatGPT-4o and ChatGPT-4.5, for comprehensive immunology research, while noting the variable capabilities of other models.

SREBP signaling: A metabolic key to B cell responses and vaccine efficacy

The SREBP case study investigated the critical role of SREBP signaling in metabolic responses to vaccination and its essential function in facilitating effective B cell responses. This study drew insights from two pivotal publications. Li *et al.* (17) demonstrated that vaccination with the Zostavax vaccine against shingles induced a transcriptional signature in the blood within 7 days of vaccination that was highly correlated with the ensuing antibody response (Fig. 4A). Contained within this signature were SREBP and its target genes. SREBP is a transcription factor that plays a key role in the orchestration of cholesterol biosynthesis (17). This raised the possibility that metabolic reprogramming, driven by SREBP, is crucial for the generation of robust immune responses after vaccination (17).

In a previous study, we investigated this by engineering mice with a targeted deletion of SCAP (SREBP cleavage-activating protein), a key regulator of SREBP signaling, in either B cells or CD11c⁺ APCs (Fig. 4B) (27). The loss of SCAP in CD11c⁺ APCs had no discernible impact on immune responses. However, SREBP signaling in B cells was essential for robust antibody production, as well as the formation of germinal centers, memory B cells, and bone marrow plasma cells (27). This pathway was also necessary for the metabolic

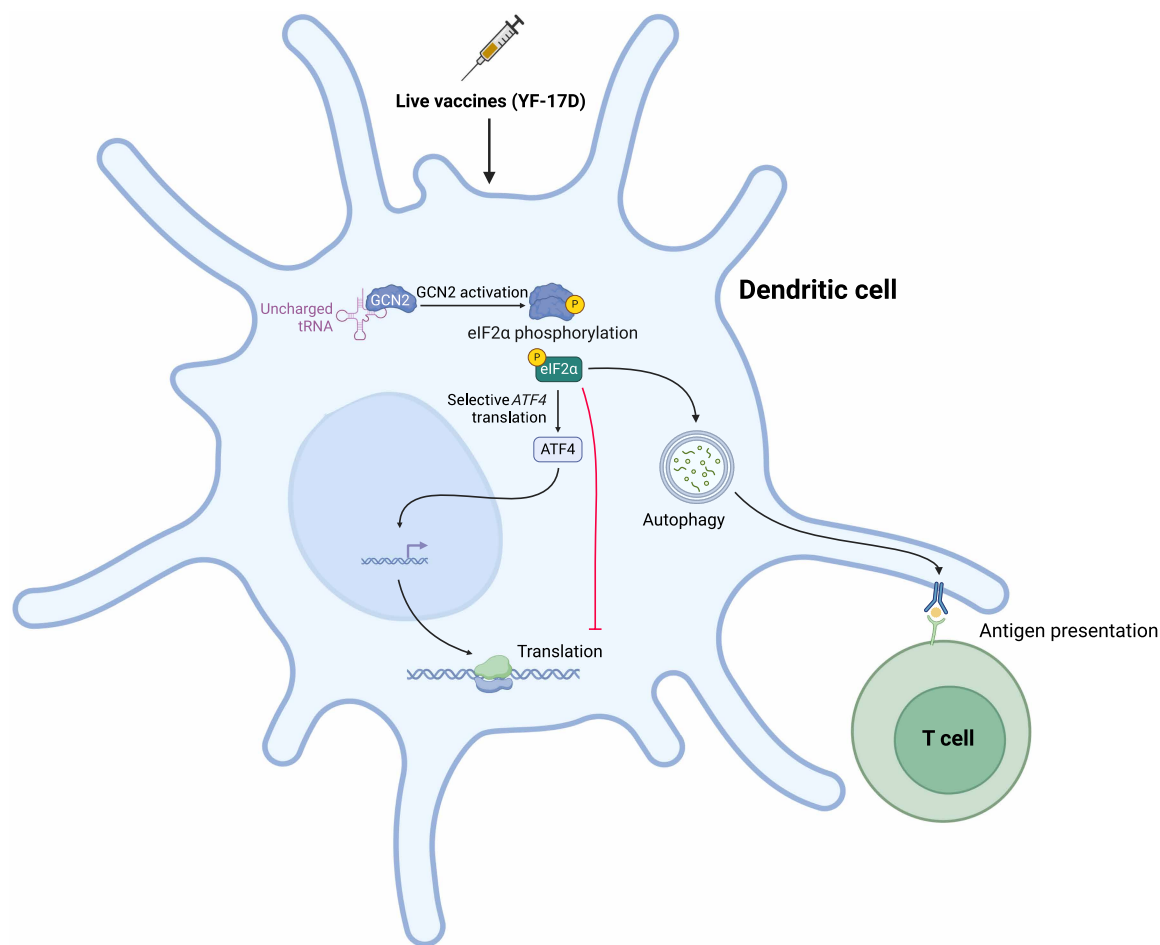


Fig. 2. GCN2-driven immune response to yellow fever vaccine YF-17D. Activation of the amino acid sensor GCN2, either by amino acid starvation or live vaccines, such as the yellow fever vaccine (YF-17D), triggers stress response pathways through eIF2α phosphorylation. This suppresses general mRNA translation but enhances the translation of specific mRNAs such as *ATF4*, a transcription factor regulating gene involved in homeostasis during nutrient stress. Phosphorylated eIF2α also induces autophagy, leading to enhanced cross-presentation of antigens, dendritic cell activation, and improved antigen presentation to CD8⁺ and CD4⁺ T cells.

	A Accurate data mining	B Formulation of biological hypothesis	C Formulation of experiments to validate hypothesis	D Broader conceptual significance of results
LLaMa-70B	• 66% validity (2/3 papers)	• GCN2 regulates immune gene translation, affecting CD8⁺ T cell response to YF	• Transfect PBMCs with GCN2 siRNA and vaccinate with YF	• Stress granules inform mRNA-based vaccine design
ChatGPT-4o	• On target reference • 100% validity (1/1 source papers)	• Post vaccine GCN2 boosts APC autophagy, antigen presentation, and CD8⁺ T cell response	• Vaccinate GCN2-deficient and wild-type mice; assess autophagy, YF-17D antigens, and CD8⁺ T cells	• GCN2 insights aid immunotherapy and vaccine design
Microsoft Co pilot	• On target reference • 100% validity (2/2 hits on same paper)	• GCN2 aids antigen presentation and T cell priming by managing immune cell stress	• Vaccinate GCN2-deficient and wild-type mice; analyze via flow cytometry and cytokine assays	• Target GCN2 to link stress response with adaptive immunity for better vaccines
SciSpace	• On target reference • 100% validity (3/3 papers)	• GCN2 boosts DC autophagy, antigen presentation, and CD8⁺ T cells	• Analyze GCN2 in PBMCs post-vaccine; test inhibitors in vitro	• Nutrient-sensing pathways shape immune responses
ChatGPT 4.5	• No references included	• GCN2 up-regulation enhances antigen presentation via metabolic and stress-response pathways, boosting vaccine-specific CD8⁺ T cell responses	• Ex vivo PBMC assays with GCN2 knockdown to assess CD8⁺ T cell responses post-antigen exposure, using flow cytometry and ISR pathway analysis	• Links metabolism to immunity, guiding biomarker use and vaccine design

Fig. 3. Summary of GCN2 case study four-tier evaluation results from five LLMs: LLaMa-70B, ChatGPT-4o, Microsoft Copilot, SciSpace, and ChatGPT-4.5. (A) The “accurate data mining” tier was evaluated on the basis of the correctness of the LLMs to find appropriate and existing references, in addition to pinpointing the historical references of the original publications for the GCN2 case study. (B) The “formulation of biological hypothesis” tier was evaluated on the basis of the interpretation of the query background given and logic of proposed hypothesis. YF, yellow fever; DC, dendritic cell. (C) The “formulation of experiments to validate hypothesis” was evaluated on the basis of validity and benefit of the proposal. siRNA, small interfering RNA; ISR, integrated stress response. (D) The “broader conceptual significance of results” was evaluated on the additional input given in comparison with historical context.

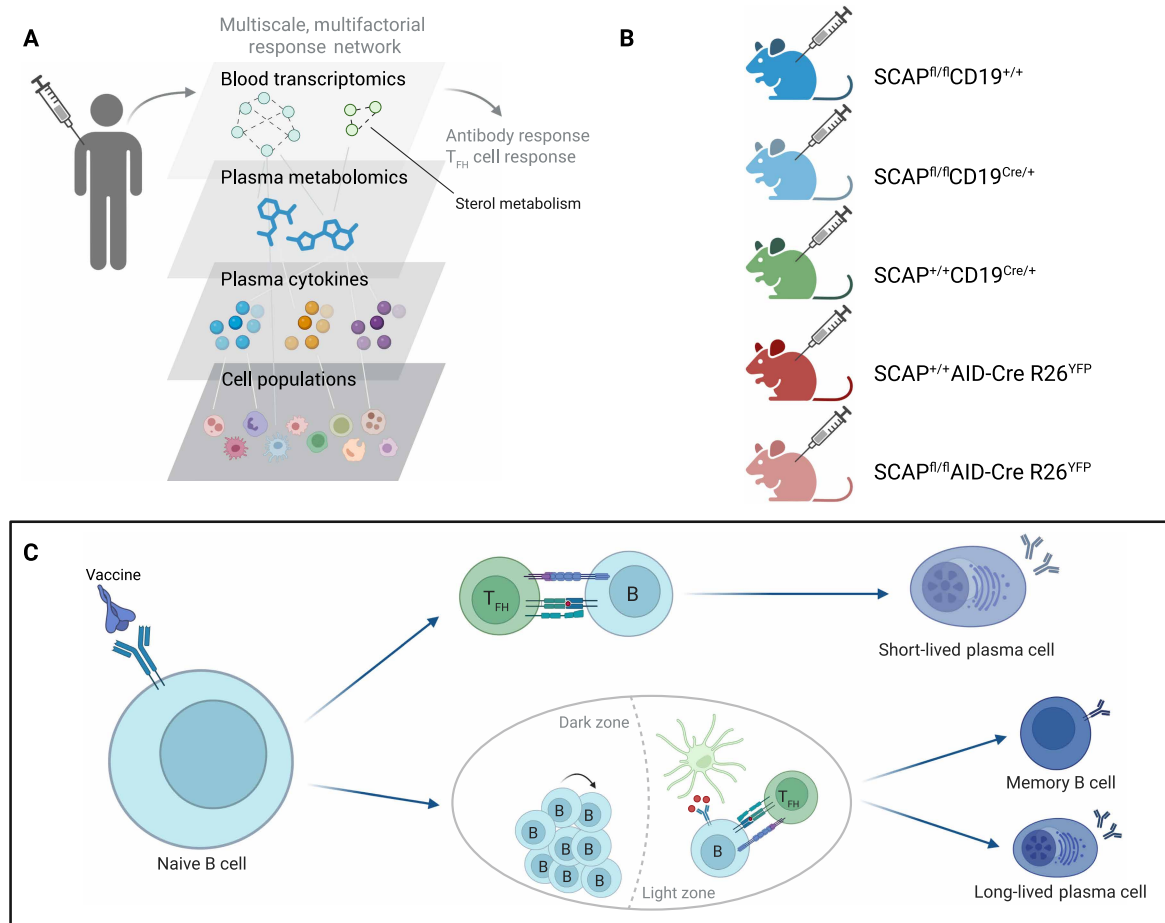


Fig. 4. SREBP signaling in B cell function and antibody responses. (A) Li *et al.* (16) found in a human clinical trial where Zostavax was administered that the *SREBF1* target network was predictive of T follicular helper cell (T_{FH} cell) response, as well as IgG response. (B) Luo *et al.* (27) compared wild-type mice and SCAP-deficient mice, highlighting the role of SCAP-dependent SREBP signaling in B cell function. They generated mice with B cell-specific SCAP deletion ($SCAP^{fl/fl}CD19^{Cre/+}$) and compared those mice with control mice ($SCAP^{fl/fl}CD19^{+/+}$ and $SCAP^{+/+}CD19^{Cre/+}$). To examine the functions of SREBP signaling in germinal center B cells, they generated $SCAP^{fl/fl}AID-Cre R26^{YFP}$ mice and compared those mice with controls ($SCAP^{+/+}AID-Cre R26^{YFP}$). (C) Antigen recognition triggers T cell activation, inducing effector molecule expression that activates B cells. This activation drives B cell differentiation, resulting in robust germinal center responses and the formation of memory B cells and antibody-secreting plasma cells.

reprogramming of activated B cells. When stimulated with mitogens, SCAP-deficient B cells failed to proliferate effectively and exhibited a reduction in lipid raft formation. Furthermore, conditional deletion of SCAP in germinal center B cells using activation-induced deaminase-Cre (AID-Cre) led to a decrease in lipid raft content and impaired cell cycle progression. These findings, which have been published previously, highlight a mechanistic link between sterol metabolism and the magnitude and persistence of humoral immune responses. Together, these studies underscore the interplay between metabolic pathways and immune responses, particularly in the context of vaccination, suggesting that enhancing SREBP signaling could bolster B cell responses and thereby improve vaccine efficacy (Fig. 4C). Conversely, modulating SREBP signaling could offer therapeutic potential for conditions driven by excessive antibody production, such as autoimmune disorders and allergic diseases. In the current study, this previously characterized pathway served as a metabolic case study to evaluate how LLMs reason about lipid metabolism and immune regulation within our four-tier framework.

Building upon these prior insights, we assessed whether LLMs could accurately recall mechanistic information about SREBP signaling, generate testable hypotheses linking sterol metabolism to antibody durability, propose experimental strategies to validate such connections, and infer broader conceptual relationships between cellular metabolism and immune responsiveness. Focusing on SREBP as a case study enabled a deeper exploration of how metabolic pathways shape the magnitude and longevity of vaccine responses and how AI-assisted reasoning might reveal previously unidentified perspectives on immunometabolic regulation relevant to both vaccine optimization and immune-mediated disease.

Performance evaluation of SREBP case study across multiple LLMs

The evaluation of the SREBP case study was conducted using five LLMs—ChatGPT-4o, ChatGPT-4.5, Microsoft Copilot, SciSpace, and LLaMA. In the main comparison (Fig. 5), ChatGPT-4o, Microsoft Copilot, and SciSpace demonstrated strong capabilities in accurate data mining, each achieving 100% validity in extracting

	A Accurate data mining	B Formulation of biological hypothesis	C Formulation of experiments to validate hypothesis	D Broader conceptual importance of results
LLaMa-70B	• 33% validity (1/3 papers)	• SREBPs regulate lipid metabolism, immune function, and antigen presentation	• Knock down SREBP in PBMCs and assess Zostavax response	• Lipid metabolism links to immune regulation and antigen processing
ChatGPT-4o	• 100% validity (1/1 source papers)	• High SREBP boosts lipid biosynthesis	• Vaccinate mice with B cell-specific SREBP deletion	• Lipid metabolism shapes immune responses
Microsoft Copilot	• 100% validity (1/1 papers and Wikipedia page)	• SREBP affects PBMC membrane fluidity and composition	• Vaccinate with Zostavax and test SREBP modulators	• Lipid metabolism regulates immunity and effects lipid composition effects
SciSpace	• 100% validity (3/3 papers)	• SREBP-driven lipid metabolism boosts antibody response	• Treat vaccinated PBMCs with SREBP modulators in a longitudinal study	• Targeting metabolic pathways may enhance vaccine efficacy
ChatGPT-4.5	• No references included	• Early SREBP up-regulation post-vaccination enhances APC function and T cell activation, boosting antibody production	• Use SREBP-modulated mouse models to assess effects on APC activity, T helper responses, and antibody titers post-vaccination	• Validating this could reveal lipid metabolism as a targetable axis for enhancing vaccine efficacy and identifying predictive biomarkers

Fig. 5. Summary of SREBP case study four-tier evaluation results from five LLMs including LLaMa-70B, ChatGPT-4o, Microsoft Copilot, SciSpace, and ChatGPT-4.5. (A) The accurate data mining tier was evaluated on the basis of the correctness of the LLMs to find appropriate and existing references, in addition to pinpointing the historical references of the original publications for the SREBP case study. (B) The formulation of biological hypothesis tier was evaluated on the basis of the interpretation of the query background given and logic of proposed hypothesis. (C) The formulation of experiments to validate hypothesis was evaluated on the basis of validity and benefit of the proposal. (D) The broader conceptional importance of results was evaluated on the additional input given in comparison with historical context.

relevant information from key literature sources (17, 27), effectively capturing the critical roles of SREBP in lipid metabolism and immune regulation. In contrast, LLaMA performed less well, correctly mining data from only one of three papers, with a 33% validity rate, indicating a need for refinement (Fig. 5A). Regarding hypothesis formulation, all models, including ChatGPT-4o, generated plausible, literature-aligned hypotheses about SREBP’s immunological functions (Fig. 5B). ChatGPT-4o suggested that increased SREBP boosts lipid biosynthesis and adaptive immunity, whereas Microsoft Copilot and SciSpace highlighted membrane composition changes and lipid metabolism’s impact on antibody responses, respectively. LLaMA linked SREBP to lipid metabolism and antigen presentation pathways. Experimental design proposals were also strong, with ChatGPT-4o recommending vaccination studies using mice with B cell-specific SREBP deletion to directly test functional effects on antibody production. Microsoft Copilot suggested Zostavax vaccination combined with SREBP modulators, SciSpace proposed longitudinal PBMC studies with SREBP activators/inhibitors, and LLaMA recommended SREBP knockdown in PBMCs to evaluate vaccine response (fig. 5C). The broader biological insights generated by these models reflected a nuanced understanding of immunometabolic regulation, with ChatGPT-4o and Microsoft Copilot emphasizing lipid metabolism’s key role in shaping immune responses and vaccine efficacy, SciSpace proposing metabolic targeting to enhance vaccine outcomes, and LLaMA recognizing the multifaceted immune functions of SREBP, including antigen processing (Fig. 5D and table S2). Meanwhile, results from ChatGPT-4.5 reinforced these findings with more detailed analyses of hypothesis formulation and experimental design, underscoring the progression in LLM capabilities (Fig. 5, B and C, and fig. S2, A and B). Overall, this evaluation highlights that ChatGPT-4o, Microsoft Copilot, and SciSpace excel in mining and integrating knowledge consistent with the literature, whereas LLaMA requires improvements. Although these models collectively support the existing understanding of SREBP in immunity, they have yet to uncover distinct mechanistic insights into the metabolic regulation of vaccine responses.

Impact of TLR5 and gut microbiota on immune responses to seasonal influenza vaccination

Our previous study of immune responses to seasonal influenza vaccination identified a strong link between expression of TLR5, a receptor for bacterial flagellin, and antibody production weeks later (14). Experiments in TLR5-deficient mice confirmed that this receptor is essential for generating vaccine-specific antibodies (36). Further, antibiotic-treated and germ-free mice lacking microbiota also showed impaired responses, suggesting that gut bacteria influence vaccine immunity (36). To test this in humans, we performed a clinical trial in which we administered broad-spectrum antibiotics before and after vaccination (28). Although antibody responses remained intact in most participants, individuals with low preexisting immunity showed reduced H1N1-specific immunoglobulin G1 (IgG1) and IgA responses (Fig. 6) (28). Antibiotic treatment also heightened inflammatory pathways [activating protein 1/nuclear receptor 4A (AP1/NR4A)], disrupted metabolic profiles, and markedly lowered serum secondary bile acids, which correlated with inflammation activation. Multiomics analysis revealed strong links between microbial species and immunometabolic signatures, underscoring the microbiome’s role in shaping vaccine responses.

The interaction between the host microbiota and the immune system represents another layer of complexity in vaccine responsiveness. Here, we leveraged this well-established pathway as an environmental-immunological model within the four-tier evaluation system to test LLM reasoning. We assessed each model’s ability to (i) recall accurate mechanistic details of TLR5 signaling and microbiota interactions, (ii) generate biologically plausible hypotheses extending these relationships, (iii) design experimental approaches for validation, and (iv) articulate broader conceptual connections between microbial ecology and vaccine-induced immunity.

Evaluation of TLR5 case study across multiple LLMs

The evaluation of the TLR5 case study across multiple LLMs revealed distinct performance patterns at various analytical tiers, particularly in the context of antibiotic-induced changes in the gut microbiome

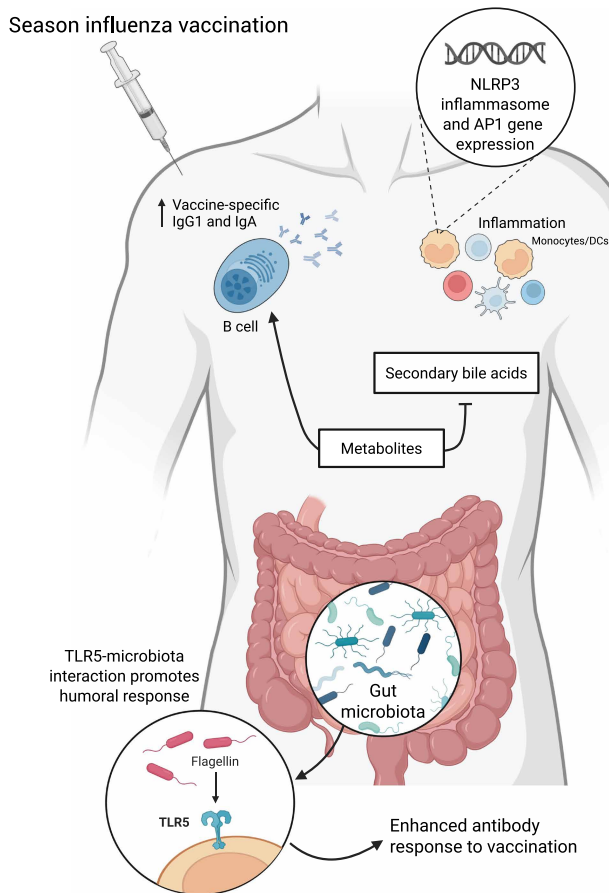


Fig. 6. Gut microbiota, inflammation, and metabolic regulation of influenza vaccine responses. In a clinical trial on seasonal influenza vaccination, Hagan *et al.* (28) found that antibiotic treatment before vaccination enhanced inflammatory signatures, including AP1/NR4A expression and dendritic cell activation, whereas altering metabolic trajectories, such as reduced serum secondary bile acids, linked to AP1/NR4A signaling and inflammasome activation. The gut microbiota support antigen-specific IgG1 and IgA responses to vaccination in healthy individuals, with secondary bile acids helping to regulate inflammation by suppressing NOD-, LRR- and pyrin domain-containing protein 3 (NLRP3) inflammasome and AP1-associated gene expression. TLR5 expressed on intestinal immune cells recognizes bacterial flagellin, linking microbiota-derived signals to systemic vaccine responses. Loss of TLR5 signaling or antibiotic depletion impairs antibody and plasma cell responses to influenza vaccination (14, 36).

and their implications for vaccine efficacy. ChatGPT-4.5's creativity assessment results are summarized in fig. S3, showcasing advanced integrative hypotheses and experimental designs that leverage spatial transcriptomics and multiomics approaches.

In parallel, a comparative assessment involving ChatGPT-4o, ChatGPT-4.5, Microsoft Copilot, SciSpace, and LLaMA is presented in Fig. 7, highlighting their varied strengths and weaknesses across four key evaluation tiers related to TLR5's role in vaccine immunity and microbiome interactions (28, 29). In terms of accurate data mining (Fig. 7A), ChatGPT-4o, Microsoft Copilot, and SciSpace excelled, each achieving 100% validity in extracting key literature findings, whereas LLaMA showed limited success with only 33% validity.

When it came to the formulation of biological hypotheses, all models provided insightful contributions. LLaMA proposed that

TLR5 boosts APC activation and B cell response to influenza vaccines, whereas ChatGPT-4o and ChatGPT-4.5 suggested that postvaccine TLR5 enhances antibody production through microbiota flagellin recognition. Microsoft Copilot emphasized TLR5's role in boosting innate immunity to enhance vaccine antibody responses, and SciSpace highlighted that TLR5 drives inflammation, which, in turn, boosts B cell activation and antibody production (Fig. 7B and fig. S3A). These hypotheses align well with the findings from the publications, which underscore the critical role of the gut microbiome in modulating immune responses to vaccines.

In terms of experimental design, the models again showcased varied levels of proficiency. LLaMA suggested testing TLR5 modulators in PBMCs with the influenza vaccine, whereas ChatGPT-4o recommended a human vaccine study involving a group receiving broad-spectrum antibiotics. Microsoft Copilot proposed a vaccination strategy involving individuals with normal and reduced TLR5 expression, and SciSpace advocated for a study examining TLR5 pre- and postvaccination using RNA sequencing and flow cytometry (Fig. 7C). These experimental designs reflect a varied understanding both positive and negative of current known experimental setups and technology.

Last, the broader conceptual importance of the results varied across the models. LLaMA noted that TLR5 shapes early immunity, APC antigen presentation, and adjuvant design. ChatGPT-4o articulated that TLR5 serves as a bridge between innate and adaptive immunity, whereas Microsoft Copilot emphasized that TLR5 modulation aids in the design of vaccines and therapies. SciSpace provided insights that guide adjuvant development for more effective vaccines (Fig. 7D). These conceptual insights resonate with the findings in the literature, which advocate for a holistic approach to vaccination that considers the state of the gut microbiome as a crucial factor influencing immunological outcomes.

Together, these analyses illustrate the complementary insights offered by different LLMs, with ChatGPT-4.5 delivering cutting-edge integrative experimental concepts (fig. S3B) and ChatGPT-4o, alongside other models, effectively mining literature and formulating hypotheses (Fig. 7). Collectively, these findings reaffirm TLR5's importance in vaccine response modulation and the pivotal influence of gut microbiota in shaping immunological outcomes (table S3).

DISCUSSION

The emergence of LLMs, such as ChatGPT-4o and ChatGPT-4.5 (OpenAI), Microsoft Copilot (Microsoft Corporation), SciSpace (Typeset.io), and LLaMA-70B (Meta Platforms), has transformed the landscape of academic research by providing powerful tools for hypothesis generation, data synthesis, and literature analysis (1–4). These models differ considerably in architecture and training scope, which likely contributed to their varied performance. ChatGPT-4o and ChatGPT-4.5, proprietary transformer-based models trained on a broad multimodal corpus, are optimized for reasoning and conversational depth. Microsoft Copilot, built on the OpenAI GPT-4 architecture but fine-tuned for productivity and integration within Microsoft 365 tools, emphasizes structured and actionable outputs. SciSpace is tailored specifically for scientific literature comprehension and citation retrieval, whereas LLaMA-70B, an earlier and comparatively smaller open-source model within Meta's expanding LLaMA family, offers transparency and adaptability but less domain-specific optimization. Given that Meta's more recent LLaMA 3 models

	A Accurate data mining	B Formulation of biological hypothesis	C Formulation of experiments to validate hypothesis	D Broader conceptual importance of results
LLaMa-70B	• 33% validity (1/3 papers)	• TLR5 boosts APC activation and B cell response to influenza vaccine	• Test TLR5 modulators in PBMCs with influenza vaccine	• TLR5 shapes early immunity, APC antigen presentation, and adjuvant design
ChatGPT-4o	• On target reference • 100% validity (1/1 source papers)	• Post vaccine TLR5 boosts antibodies via microbiota flagellin recognition	• Human vaccine study with one group receiving broad-spectrum antibiotics	• TLR5 bridges innate and adaptive immunity
Microsoft Copilot	• 100% validity (2/2 source papers and Protein Atlas webpage)	• TLR5 boosts innate immunity, enhancing vaccine antibody responses	• Vaccinate participants with normal and reduced TLR5 expression	• TLR5 modulation aids vaccine and therapy design
SciSpace	• 100% validity (5/5 papers)	• TLR5 drives inflammation, boosting B cell activation and antibodies	• Study TLR5 pre/post vaccine using RNA-seq and flow cytometry	• TLR5 insights guide adjuvant development for better vaccines
ChatGPT-4.5	• No references included	• TLR5 induction reflects enhanced innate activation (via flagellin or similar pathways) that primes adaptive immunity and improves antibody production	• Test antibody responses in models receiving influenza vaccine with or without TLR5 agonists or inhibitors, assessing innate and adaptive markers	• Validates PRR-specific innate-adaptive linkage and supports using targeted innate modulation (e.g., TLR5) to enhance vaccine design and efficacy

Fig. 7. Summary of TLR5 case study four-tier evaluation results from five LLMs including LLaMa-70B, ChatGPT-4o, Microsoft Copilot, SciSpace, and ChatGPT-4.5. (A) The accurate data mining tier was evaluated on the basis of the correctness of the LLMs in finding appropriate and existing references, in addition to pinpointing the historical references of the original publications for the TLR5 case study. (B) The formulation of biological hypothesis tier was evaluated on the basis of the interpretation of the query background given and logic of proposed hypothesis. (C) The formulation of experiments to validate hypothesis was evaluated on the basis of validity and benefit of the proposal. RNA-seq, RNA sequencing. (D) The broader conceptional importance of results was evaluated on the additional input given in comparison with historical context. PRR, pattern recognition receptor.

surpass hundreds of billions of parameters, the relatively limited scale of LLaMA-70B may explain its reduced contextual precision compared with newer, larger systems. Together, these models represent a spectrum of design priorities from general reasoning to targeted scientific synthesis, which helps explain the differences observed in their cognitive and contextual performance.

In the context of systems vaccinology, all models demonstrated aptitude for factual recall, pattern recognition, and structured hypothesis formulation. Across the three case studies, GCN2, SREBP, and TLR5, they successfully retrieved known mechanisms in immune regulation, formulated coherent hypotheses, and suggested plausible experiments, underscoring their utility for systems-level analysis. ChatGPT-4.5 showed the highest level of integrative reasoning, connecting metabolic, microbial, and immunological processes with conceptual coherence. SciSpace was effective at targeted literature retrieval but limited in mechanistic depth, whereas LLaMA-70B provided broader but less detailed responses, reflecting its generalist architecture. Despite these capabilities, the models did not generate hypotheses or experimental proposals sufficiently insightful to warrant empirical testing. Their outputs largely reorganized established findings or restated well-known mechanisms, reflecting their reliance on existing literature rather than discovery-driven inference. This pattern highlights a central constraint of current LLMs: Their strength lies in scaffolding and synthesis, not in autonomous hypothesis generation or risk-taking. These findings mirror broader challenges in AI-assisted research, including the tendency of models to reinforce prevailing paradigms and create the illusion of understanding, where outputs appear coherent yet mask superficial comprehension (11). Such limitations reaffirm the importance of human interpretation and critical oversight to ensure that AI remains a catalyst for scientific inquiry rather than a constraint on creativity.

A related epistemological challenge is the difficulty of distinguishing synthesis from reasoning. Because these models are trained on the same literature they analyze, their responses can reflect subtle reproductions of existing knowledge rather than independent conceptual inference. This phenomenon complicates assessments of model “understanding” and underscores the need for objective

measures of conceptual distance and interpretive depth. Nevertheless, the LLMs exhibited proficiency in identifying mechanistic patterns and proposing structured lines of inquiry. Comparable advances in symbolic and structured reasoning in fields such as mathematics, exemplified by AlphaGeometry2 (37), suggest that combining LLMs with algorithmic reasoning frameworks may improve their ability to formulate biologically grounded insights. Such integration could enhance model transparency, logical consistency, and cross-domain reasoning in future iterations.

Several methodological limitations also warrant consideration. Our four-tier evaluation framework, which encompassed factual accuracy, hypothesis generation, experimental design, and conceptual synthesis, relied on qualitative assessment by expert reviewers. This introduces potential subjectivity, given that no inter-rater reliability or quantitative distance metrics were applied. In addition, data overlap between training and evaluation corpora could have influenced model performance, and the use of single-shot prompts does not replicate the iterative dialogue typical of human research workflows. Future assessments should incorporate multiturn prompting and chain-of-thought reasoning to better simulate real-world scientific engagement. Moreover, our focus on text-based reasoning excluded the rapidly advancing multimodal models that integrate textual, molecular, and imaging data, which may be better suited for analyzing complex biological systems.

Despite these constraints, this study illustrates the growing promise of LLMs as augmentative tools for scientific reasoning. They can accelerate literature synthesis, connect disparate biological findings, and support hypothesis refinement, enhancing the efficiency and breadth of discovery efforts. As next-generation models evolve, integrating structured reasoning algorithms, domain-specific fine-tuning, and transparent data provenance could expand their interpretative capacity. However, their effective use will depend on maintaining rigorous validation, ethical governance, and human oversight. The most constructive path forward is a partnership between human expertise and AI analysis, cultivating hybrid intelligence that combines computational power with the interpretive, critical, and ethical strengths of scientific researchers.

MATERIALS AND METHODS

Study design

This study was designed to systematically evaluate the cognitive and scientific reasoning capabilities of five LLMs using a standardized, reproducible framework. The primary objective was to assess the ability of each model to perform core components of scientific inquiry, data mining, hypothesis generation, experimental design, and conceptual synthesis, within the domain of systems vaccinology. Three mechanistically distinct, experimentally validated case studies (GCN2, SREBP, and TLR5) were selected as biological test scenarios. Because this was a computational assessment, no human participants or animal subjects were involved, and traditional variables such as sample size, randomization, or blinding were not applicable. Each model was queried independently under identical conditions, and all responses were archived and analyzed for accuracy, coherence, and biological plausibility. This study adheres to the EQUATOR Network's Guidelines for Reporting guideline for the use of generative AI tools in medical research: the GAMER statement to ensure reproducibility and transparency.

LLM model selection

Five LLMs were selected on the basis of their prominence in research, accessibility, and architectural diversity: ChatGPT-4o and ChatGPT-4.5 (OpenAI, San Francisco, CA, USA), Microsoft Copilot (Microsoft Corporation, Redmond, WA, USA), LLaMA-70B (Meta Platforms Inc., Menlo Park, CA, USA), and SciSpace (Type-set.io, Bengaluru, India). Model interactions with ChatGPT-4o, SciSpace, Microsoft Copilot, and LLaMA-70B were performed between 11 and 13 December 2024, whereas ChatGPT-4.5 analyses were conducted between 4 and 5 March 2025, after its public release. These dates are reported to document the versioning context for reproducibility.

Differences between ChatGPT-4o and ChatGPT-4.5 are as follows: ChatGPT-4o is a multimodal model capable of processing text, image, and audio inputs in real time, optimized for balanced speed and contextual reasoning. ChatGPT-4.5 represents an enhanced version with extended context length, refined reasoning algorithms, and improved factual grounding. Whereas 4o was included for its real-time adaptability and general accessibility, 4.5 was specifically tested for its deeper analytical reasoning and contextual synthesis, particularly within complex biological frameworks.

Selection criteria were as follows:

1) Model availability and accessibility: All models were publicly available through official web or institutional platforms, ensuring reproducibility.

2) Training data and specialization: ChatGPT and Microsoft Copilot were trained as generalist systems using broad corpora, whereas SciSpace was trained on scientific literature and metadata, and LLaMA-70B represented an open-source alternative designed for research transparency.

3) Performance in scientific inquiry: Prior reports and anecdotal evidence indicated that these models are frequently used for literature synthesis, hypothesis formulation, and data interpretation in scientific workflows.

4) Interface and customization: Models differed in interaction structure from guided scientific summarization (SciSpace) to conversational reasoning (ChatGPT, Copilot, and LLaMA), allowing comparison of how interface design influences reasoning output.

Evaluation framework

Each model was evaluated across four standardized tiers:

- 1) Data mining accuracy
- 2) Hypothesis generation
- 3) Experimental design
- 4) Conceptual reasoning or synthesis

Three biological case studies, GCN2 (metabolic stress and antigen presentation), SREBP (lipid metabolism and B cell responses), and TLR5 (microbiota and innate immune signaling), were used to assess model reasoning across cellular, metabolic, and environmental dimensions of immunity.

Data recording and comparison

All model interactions were conducted in separate sessions to avoid data contamination between prompts. Responses were exported in plain-text format. Each model received identical base queries to ensure comparability, and exploratory follow-up prompts were used only where applicable. Outputs were evaluated independently by two reviewers with immunology expertise, who assessed each response on factual accuracy, scientific coherence, biological plausibility, and creativity of reasoning using a structured rubric. Discrepancies in assessment were resolved by consensus. The model outputs are available on the Zenodo platform (DOI: 10.5281/zenodo.15531315).

The rationale for dual prompting with ChatGPT-4.5 is that it was selected for additional dual prompting experiments due to its higher reasoning consistency and longer context capacity compared with other models. Two prompting modes were used:

- 1) Canonical prompts grounded in existing literature to test factual recall and standard reasoning.
- 2) Exploratory prompts designed to probe the model's capacity for cross-domain association and counterfactual reasoning.

For instance, the canonical query "We have defined a gene expression signature in which the expression of the gene for GCN2 in PBMCs, within a day of vaccination, correlates with the later CD8⁺ T cell response to yellow fever YF-17D vaccination. Could you: (i) identify relevant references, (ii) propose a hypothesis to explain this result, (iii) suggest an experiment to validate it, and (iv) infer broader biological implications?" was contrasted with exploratory prompts such as "What hypotheses about this transcriptional signature are biologically plausible but underexplored in current literature?" and "If you were designing a new experiment in this area, what approach might reveal mechanisms not yet studied?" This comparative prompting strategy was intended to evaluate whether guided reasoning could enhance conceptual depth beyond factual synthesis.

Replication and data transparency

Each prompt was tested in triplicate per model to account for response variability, and outputs were stored verbatim. Because model responses may vary across runs, this replication ensured stability of conclusions across trials. All data were analyzed qualitatively, and no data were excluded. There were no outliers or stopping rules, as the analysis was deterministic within a closed computational framework.

Supplementary Materials

The PDF file includes:

Figs. S1 to S3

Tables S1 to S3

Other Supplementary Material for this manuscript includes the following:

MDAR Reproducibility Checklist

REFERENCES AND NOTES

- OpenAI, <https://openai.com/>.
- Microsoft Copilot, <https://copilot.microsoft.com/>.
- SciSpace AI Research Agent, <https://scispace.com/>.
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results from all evaluated LLMs are included within Zenodo platform (doi: 10.5281/zenodo.15531315). Intermediary query results from all evaluated LLMs are included as supplementary tables.

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Assessing AI's cognitive abilities for scientific discovery in the field of systems vaccinology

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Editor's summary

Immunology is a field that has been inundated with dense datasets and an ever-expanding body of literature, so large language model (LLM)–based artificial intelligence applications have the potential to enhance discovery. Rodriguez-Coffinet *et al.* describe an approach, called “The Creation Game,” as a tool to compare five different LLMs for their ability advance immunology research. This framework tests LLMs for their abilities to recall relevant literature, formulate hypotheses, design experiments, and infer broader conceptual significance. Three case studies in systems vaccinology were evaluated, which revealed the relative strength of LLMs in recalling facts and detecting patterns. By contrast, LLMs were less effective in generating novel hypotheses and proposing experiments that could lead to conceptual advances. These findings highlight aspects of LLMs that can support scientific discovery in partnership with human expertise. —Christiana N. Fogg

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